

Integrated public-pilot evidence matrix

	TCGA bulk	Spatial ST	scRNA/cell-type	Drug/perturbation	Experimental validation	Claim strength
RAI differentiation / redifferentiation axis	Strong	Strong	Moderate	Moderate	Strong	Moderate
HLA/APM immune visibility axis	Strong	Strong	Moderate	Weak	Strong	Moderate
HLA-low invisible tumor axis	Strong	Moderate	Weak	Weak	Strong	Moderate
CD8 exclusion / myeloid-CAF barrier	Moderate	Strong	Moderate	Not tested	Strong	Moderate
Drug-delivery failure proxy	Moderate	Moderate	Not tested	Not tested	Strong	Weak
Aggressive dediff. / proliferation axis	Strong	Moderate	Weak	Moderate	Strong	Moderate
AI companion diagnostic feasibility	Strong	Strong	Moderate	Weak	Strong	Strong
Drug/perturbation candidate feasibility	Weak	Weak	Not tested	Moderate	Strong	Moderate

Categories summarize proposal-readiness of support; public data remain hypothesis-generating.